

FORMULAS · WORKED EXAMPLES · HEALTH CHECK

Referral Program & Google Ads Economics, Explained

Every formula, a full worked example on both channels, and a margin-sensitivity check on whether the referral program and Google Ads are actually healthy, not just profitable this month.

1. Referral Program Formulas — Explained
2. Google Ads Formulas — Explained
3. Worked Examples — Referral Program & Google Ads
4. What Peter Should Actually Be Looking At
5. Is Google Ads Still Healthy, or Eating Away Margins?
6. How to Calculate Your Real Numbers
7. Sensitivity Analysis Across Different Margins
8. Health Status & Bottom Line

1. Referral Program Formulas — Explained

$$\text{Total Acquisition Cost} = \text{Referrer Commission} + \text{Customer Discount}$$

This is everything you give up to win one referred install, the cash paid to the referrer plus the revenue you forgo through the customer's discount. Both are real costs, not just the commission line.

$$\text{Referrer Commission} = \text{Installation Value} \times \text{Commission \%}$$

What you pay the person who referred the deal, calculated straight off the contract value.

$$\text{Customer Discount} = \text{Installation Value} \times \text{Discount \%}$$

What the referred customer saves off their invoice, forgone revenue on your side.

$$\text{CAC \% of Deal Value} = \text{Total Acquisition Cost} \div \text{Installation Value}$$

Tells you what share of each deal's revenue is spent just acquiring it. Useful for a quick gut check, but revenue-based, not profit-based, so it can look deceptively small.

$$\text{Gross Profit} = \text{Revenue} - \text{COGS}$$

What's left after the direct cost of materials, labor, and logistics for that install, before touching overhead or acquisition costs.

$$\text{Gross Margin \%} = \text{Gross Profit} \div \text{Revenue}$$

The percentage of each shilling of revenue that's actually profit before overhead.

$$\text{CAC-to-Margin Ratio} = \text{Total Acquisition Cost} \div \text{Gross Profit}$$

The real safety check. Instead of asking “what % of revenue did acquisition cost,” this asks “what % of my actual profit did acquisition cost consume.” This is the number that tells you if a channel is quietly draining the business, even while revenue looks fine.

$$\text{Contribution Margin} = \text{Gross Profit} - \text{Total Acquisition Cost}$$

What's genuinely left over per install after paying for the referral. If this shrinks toward zero or goes negative, the program is costing more than it's contributing.

$$\text{Minimum Margin Required for Safety} = \text{CAC \%} \div \text{Target CAC-to-Margin Ratio}$$

Works backward from your CAC % to tell you the minimum gross margin you need to keep the ratio at a safe level (0.35). If your actual margin is below this number, no amount of volume fixes it, only better margins or renegotiated terms do.

$$\text{Program ROI} = (\text{Total Gross Profit} - \text{Total Acquisition Cost}) \div \text{Total Acquisition Cost}$$

For every shilling spent acquiring referrals, how many shillings of profit came back.

2. Google Ads Formulas — Explained

$$\text{CTR (Click-Through Rate)} = \text{Clicks} \div \text{Impressions}$$

How often people who see your ad actually click it, a signal of how relevant and compelling the ad is.

$$\text{CPC (Cost Per Click)} = \text{Ad Spend} \div \text{Clicks}$$

What you pay, on average, for each click, driven by competition for the keyword and your ad quality.

$$\text{Conversion Rate} = \text{Leads} \div \text{Clicks}$$

Of the people who clicked, how many actually became a lead (call, form, WhatsApp message).

$$\text{Cost Per Lead} = \text{Ad Spend} \div \text{Leads}$$

What it costs you to generate one lead, the number most useful for budgeting.

$$\text{CAC (Ads)} = \text{Ad Spend} \div \text{Installs Closed}$$

What it actually costs to win one paying customer through ads, the number that matters more than cost per lead, since not every lead converts.

$$\text{ROAS} = \text{Revenue Generated} \div \text{Ad Spend}$$

For every shilling spent, how many shillings came back in revenue. Your current campaign runs at 40x.

$$\text{ROI} = (\text{Revenue Generated} - \text{Ad Spend}) \div \text{Ad Spend}$$

Same idea as ROAS but expressed as net return rather than a multiple, useful for comparing against other investment types.

3. Worked Examples

Referral Program (17 Referrals)

Tier 1: 150,000–250,000 (7 referrals, assumed avg. KES 180,000)

Metric	Formula	Per Install	× 7
Commission (5%)	$180,000 \times 5\%$	KES 9,000	KES 63,000
Discount (3%)	$180,000 \times 3\%$	KES 5,400	KES 37,800
Total Acquisition Cost	$9,000 + 5,400$	KES 14,400	KES 100,800
Revenue	–	KES 180,000	KES 1,260,000

Tier 2: Above 250,000 (10 referrals, assumed avg. KES 383,000)

Metric	Formula	Per Install	× 10
Commission (5%)	$383,000 \times 5\%$	KES 19,150	KES 191,500
Discount (4%)	$383,000 \times 4\%$	KES 15,320	KES 153,200
Total Acquisition Cost	$19,150 + 15,320$	KES 34,470	KES 344,700
Revenue	–	KES 383,000	KES 3,830,000

Combined (17 referrals)

Metric	Formula	Result
Total Revenue	$1,260,000 + 3,830,000$	KES 5,090,000
Total Commissions	$63,000 + 191,500$	KES 254,500
Total Discounts	$37,800 + 153,200$	KES 191,000
Total Acquisition Cost	$254,500 + 191,000$	KES 445,500
Blended CAC % of Revenue	$445,500 \div 5,090,000$	8.75%

At an assumed 25% gross margin

Metric	Formula	Result
Total Gross Profit	$5,090,000 \times 25\%$	KES 1,272,500
CAC-to-Margin Ratio	$445,500 \div 1,272,500$	0.35
Program ROI	$(1,272,500 - 445,500) \div 445,500$	1.86

Minimum margin needed to stay safe:

- Tier 1: $8\% \div 0.35 = 22.9\%$ minimum gross margin

- Tier 2: $9\% \div 0.35 = 25.7\%$ minimum gross margin

Google Ads

KES 12,914 spent over 24 days, KES 517,000 revenue generated.

Metric	Formula	Result
ROAS	$517,000 \div 12,914$	40.0x
ROI	$(517,000 - 12,914) \div 12,914$	3,904%

4. What Peter Should Actually Be Looking At

It's easy to get pulled toward the raw profit number, “we made KES 80,000 this month,” because it feels concrete and satisfying. But a raw profit figure on its own tells you almost nothing about whether the business is healthy, since it doesn't say what it cost to get there or whether that cost is climbing.

The three numbers that actually matter, in order of importance:

- 1 CAC-to-Margin Ratio (both channels)** — this is the single most important number in this whole document. It answers “how much of what I earned did I have to give away to earn it,” which is the real test of whether a channel is sustainable, not whether it made money this month.
- 2 Trend over time, not a single snapshot** — one month's ratio means little. What matters is whether the ratio is climbing (getting worse as you scale) or holding steady/improving. A KES 80,000 profit month that came with a 0.25 ratio is genuinely healthy; the same KES 80,000 with a 0.42 ratio is a warning sign wearing a good-news number.
- 3 Margin %, not margin KES** — KES profit numbers grow naturally as revenue grows, even if the business is getting less efficient. The percentage is what tells you if efficiency is actually improving or just being masked by higher volume.

The habit to build: Every time a profit figure comes up in conversation, immediately ask “at what CAC-to-margin ratio, and compared to last month?” A shilling figure without that context is close to meaningless for decision-making, it's a headline, not an answer.

5. Is Google Ads Still Healthy, or Eating Away Margins?

So far, ROAS and ROI have been used to judge the ads channel, and both look excellent (40x, 3,904%). But ROAS is a revenue measure, it doesn't account for your actual margin, the same blind spot that made the referral program look deceptively safe until the CAC-to-margin ratio was applied. The same check needs to run on ads.

Ad CAC-to-Margin Ratio = Total Ad Spend $\div$ Gross Profit Generated from Ad-Driven Installs

This is the direct equivalent of the referral program's safety check, applied to ads instead.

Worked example, using the KES 517,000 in ad-driven revenue and the same 25% assumed margin used throughout this document

Metric	Formula	Result
Ad-Driven Gross Profit	$517,000 \times 25\%$	KES 129,250
Ad CAC-to-Margin Ratio	$12,914 \div 129,250$	0.10

At 0.10, Google Ads sits comfortably in the healthy range, far below the 0.30 threshold, meaning ads are consuming only about 10% of the profit they generate, versus the referral program's 35%. This is the number that actually confirms ads are the stronger channel, not just the ROAS multiple.

Signals that Google Ads is starting to eat into margins instead of building them:

- ◆ **Rising CPC without a matching rise in conversion rate** — you're paying more per click but not converting more of them, a sign of increased keyword competition eating efficiency.
- ◆ **Ad CAC-to-Margin Ratio climbing month over month** — even if ROAS still looks good, a climbing ratio means margin is quietly being consumed faster than revenue is growing.
- ◆ **Cost Per Lead rising faster than close rate improves** — if leads get more expensive but don't convert better, CAC per install rises even if raw lead volume looks fine.
- ◆ **ROAS holding steady while gross margin % declines** — this can happen if material/labor costs rise while ad performance stays flat; ROAS alone won't catch it, only tracking margin separately will.

6. How to Calculate Your Real Numbers

Worked examples for the four inputs this whole document currently assumes rather than measures.

1 Real Gross Margin

$$\text{Gross Margin \%} = (\text{Revenue} - \text{COGS}) \div \text{Revenue}$$

Example, using placeholder costs for a KES 180,000 install

Revenue	KES 180,000
Materials	KES 95,000
Direct Labor	KES 25,000
Logistics/Site Visit	KES 8,000
COGS Total	$95,000 + 25,000 + 8,000 = \text{KES } 128,000$
Gross Profit	$180,000 - 128,000 = \text{KES } 52,000$
Gross Margin %	$52,000 \div 180,000 = 28.9\%$

Do this for a handful of real completed installs (ideally one from each tier), then average the result, that average replaces the 25% placeholder used everywhere else in this document.

2 Close Rate (Leads → Installs)

$$\text{Close Rate} = \text{Installs Closed} \div \text{Total Leads Received}$$

Example, using placeholder monthly data

Leads received in a month	40
Installs closed from those leads	8
Close Rate	$8 \div 40 = 20\%$

Pull this from your actual lead log (Google Ads leads + referral leads combined, or split by channel for more precision) over the last 2-3 months for a reliable figure, one month alone can be noisy.

3 Ad CAC Using Real Installs Won

$$\text{CAC (Ads)} = \text{Ad Spend} \div \text{Installs Closed from That Spend}$$

Example, using the actual KES 12,914 spend with a placeholder install count

Ad Spend	KES 12,914
Installs Closed from this campaign	3 (placeholder, needs confirming)
CAC	$12,914 \div 3 = \text{KES } 4,305 \text{ per install}$

Compare this CAC against the gross profit per install (from the margin calculation above) to see the real CAC-to-margin ratio for ads, not just the revenue-based ROAS figure.

4 Overhead Allocation for the Referral Program

$$\text{Overhead Per Install} = \text{Total Program Overhead} \div \text{Number of Installs in the Period}$$

Example, using placeholder monthly costs

Peter's admin time on referrals	$\sim 6 \text{ hours/month} \times \text{KES } 1,000/\text{hour} = \text{KES } 6,000$
Landing page hosting/maintenance	KES 2,000/month
WhatsApp/tracking tools	KES 1,000/month
Total Monthly Overhead	KES 9,000
Installs that month	3
Overhead Per Install	$9,000 \div 3 = \text{KES } 3,000$

Add this KES 3,000 to the Total Acquisition Cost formula for a more accurate (and slightly higher) CAC-to-margin ratio than the commission + discount-only figures used earlier in this document.

7. Sensitivity Analysis: CAC-to-Margin Ratio Across Different Margins

Since every calculation in this document depends on the 25% margin assumption, here's how the referral program and Google Ads both look across a realistic range, so the analysis isn't fragile to one guess being wrong.

Referral Program — Tier 1 (150k–250k, CAC % = 8%)

Gross Margin %	Gross Profit (on KES 180,000)	Total Acquisition Cost	CAC-to-Margin Ratio	Status
15%	KES 27,000	KES 14,400	0.53	Caution
20%	KES 36,000	KES 14,400	0.40	Caution
25%	KES 45,000	KES 14,400	0.32	Needs Attention
30%	KES 54,000	KES 14,400	0.27	Healthy
35%	KES 63,000	KES 14,400	0.23	Healthy

Referral Program — Tier 2 (Above 250k, CAC % = 9%)

Gross Margin %	Gross Profit (on KES 383,000)	Total Acquisition Cost	CAC-to-Margin Ratio	Status
15%	KES 57,450	KES 34,470	0.60	Caution
20%	KES 76,600	KES 34,470	0.45	Caution
25%	KES 95,750	KES 34,470	0.36	Needs Attention
30%	KES 114,900	KES 34,470	0.30	Needs Attention (edge)
35%	KES 134,050	KES 34,470	0.26	Healthy

Google Ads (CAC = KES 12,914 total spend)

Gross Margin %	Gross Profit (on KES 517,000)	Ad Spend	CAC-to-Margin Ratio	Status
15%	KES 77,550	KES 12,914	0.17	Healthy
20%	KES 103,400	KES 12,914	0.12	Healthy
25%	KES 129,250	KES 12,914	0.10	Healthy
30%	KES 155,100	KES 12,914	0.08	Healthy
35%	KES 180,950	KES 12,914	0.07	Healthy

What the Sensitivity Analysis Shows

- ◆ The referral program's safety is entirely margin-dependent. Below roughly 27–30% margin, Tier 1 tips into caution territory. Below roughly 33–35%, Tier 2 does too. If G&P's real margins sit in the low-to-mid 20s, which is common for hardware-heavy installation businesses, the referral program is very likely already in the red zone, not just “needs attention.”
- ◆ Google Ads stays healthy across the entire realistic margin range. Even at a pessimistic 15% margin, ads sit at 0.17, comfortably under the 0.30 threshold. This is the strongest evidence in the whole document that ads are structurally safer, it holds up regardless of which margin assumption turns out to be true.
- ◆ The single most urgent number to get from Peter is the real margin. Everything else in this document is secondary to that one figure, it's the swing variable that determines whether the referral program is fine or actively losing money.

8. Health Status: Healthy / Needs Attention / Caution

Status	CAC-to-Margin Ratio	ROAS	What It Means
Healthy	Below 0.30	Above 20x	Channel is efficient, safe to scale spend or volume with confidence
Needs Attention	0.30–0.40	10x–20x	Still profitable, but margin is getting consumed faster than ideal, monitor closely, avoid expanding terms further, look for ways to tighten costs or renegotiate
Caution	Above 0.40	Below 10x	Channel is consuming too much profit relative to what it returns, pause scaling, revisit structure or spend allocation before committing more resources

Where both channels currently sit:

Referral Program: **Needs Attention**

CAC-to-margin ratio of 0.35 (at the 25% margin assumption) sits right in the middle of the caution band, profitable, but with limited room before it tips into unsafe territory. This is exactly why the personal-install request and elevated discount for Murage deserve scrutiny before agreeing, they'd push this ratio higher, not lower.

Google Ads: **Healthy**

Ad CAC-to-Margin Ratio of 0.10, far below the 0.30 threshold, confirms ads aren't just outperforming on ROAS (40x), they're also structurally far safer on margin exposure than the referral program currently is.

BOTTOM LINE

That combination, strong return and low margin risk, is exactly why Google Ads should carry the scaling weight while the referral program stays in “monitor and negotiate” mode.